

4.Artificial Neural Network

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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

ann = MLPClassifier(
    hidden_layer_sizes=(5,),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
)

ann.fit(X_train, y_train)

y_pred = ann.predict(X_test)

print("Actual Classes:")
print(y_test)

print("\nPredicted Classes:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy:
1.0

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Classification Report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00        19
     1       1.00      1.00      1.00        13
     2       1.00      1.00      1.00        13

 accuracy          1.00
 macro avg          1.00
weighted avg          1.00
```